# Computer Systems Servicing

# 11

# Quarter 2

# Self-Learning Module 1

# Computer Operating System


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# EXPECTATIONS

A. define computer operating systems;
B. discuss the history of computer operating systems;
C. appreciate the importance of computer operating systems.

# PRETEST

**Direction:** Identify what is referred to in the following statements. Write your answers on the lines provided.

_____1. It manages the computer's memory, processes, and all its software and hardware.
_____2. Created the Windows operating system in the mid-1980s.
_____3. The first independent version of Microsoft Windows, released on November 20, 1985, achieved little popularity.
_____4. A successor to Windows 98 called Windows ME, short for "Millennium Edition". It was the last DOS-based operating system from Microsoft.
_____5. A line of operating systems created by Apple. It comes preloaded on all new Macintosh computers, or Macs.
_____6. Released on July 20, 2011, it brought developments made in Apple's iOS, such as an easily navigable display of installed applications (Launchpad) and (a greater use of) multi-touch gestures, to the Mac.
_____7. A family of open source operating systems, which means that they can be modified and distributed by anyone around the world.
_____8. an operating system based on Arch Linux, which is fast, user-friendly and desktop oriented.
_____9. It started life under the name Cinnarch, merging the Cinnamon desktop with the Arch Linux distribution.
_____10. A full Linux desktop operating system, available free of charge with community and technical support.

# RECAP

In the previous module, you have learned how to creating image file/s of installers from different media devices using Power ISO. You were made aware that image file is composed of multimedia data in one of these formats to raster data for screen display or printer use. PowerISO is a powerful CD / DVD / BD image file processing tool for opening, extracting, burning, creating, editing, compressing, encrypting, splitting, and converting ISO files with an internal virtual drive. Fast all CD / DVD / BD image files including ISO and BIN files can be processed. PowerISO arrives with an all-in-one approach. With these, you can create image files of installers from different media devices using the software Power ISO. This module is the first topic in the second quarter where you will know about the operating systems.

# LESSON

**Operating system** is a widely generalized device program, or "**OS**." The most important software which runs on a computer is an operating system. This manages the memory, operations, and all of the computer's applications and hardware. It also allows you to communicate with the computer, without knowing how to speak the "language" of the computer. A computer is useless without an operating system.

Operating system (OS) of your computer manages all of the computer's software and hardware. Most of the time, there are several different computer programs running simultaneously and all of them need to access the central processing unit (CPU), memory, and storage of your computer. All of this is managed by the operating system to ensure each application gets what it wants.

**TYPES OF OPERATING SYSTEMS**

The three most common operating systems for personal computers are:
**Microsoft Windows, Apple Mac OS X, and Linux.**

***Image 1.1*** *From* Left to Right Windows Logo, Linux Logo, Mac OS

## ➢ Microsoft Windows

Microsoft created mid-1980s Windows operating system. There have been many different Windows versions over the years, but the latest are Windows 10 (released in 2015), Windows 8 (2012), Windows 7 (2009) and Windows Vista (2007). Windows comes preloaded to most new PCs, helping to make it the world's most popular operating system.

## History of Windows OS

**Windows 1.0-** Microsoft Windows' first standalone version, version 1.0, released on 20 November 1985, achieved little attention.

**Windows 2.0 -** On December 9, 1987, Microsoft Windows version 2 came out and became marginally more successful than its predecessor.

**Windows 3.0-** Released in May 1990, improved native-applications capabilities. This also allowed users to boost multitasking with older MS-DOS based applications relative to Windows/386, due to the virtual memory implementation.

**OS/2 -** Microsoft and IBM had cooperatively developed OS/2 as a successor to DOS during the mid to late 1980s. OS/2 1.0, released in 1987, included support for switching and multitasking, and allowed DOS executables to run.

**Windows 3.1 -** Microsoft developed Windows 3.1 (first launched in April 1992) in reaction to the imminent arrival of OS/2 2.0, which included many enhancements to Windows 3.0, such as viewing TrueType portable fonts (built together with Apple), enhancing disk output in 386 Enhanced Mode, multimedia support, and bugfixes.

**Windows NT -** A family of Microsoft-produced operating systems, the first version of which came out in July 1993. It is a multi-user, multiprocessing, processor-independent operating system. Windows NT's first version was Windows NT 3.1, which was designed for workstations which server computers.

**Windows 95 -** Codenamed Chicago) is a consumer-oriented Microsoft Operating System. It was released on August 24, 1995, and marked a major change over previous DOS-based Windows devices from the company.

**Windows NT 4.0 -** The successor to Windows NT 4.0 was 3.51 (1995), and 3.5 (2009). In July 1996, one year after the release of Windows 95, Microsoft released Windows NT 4.0 for development.

**Windows 98 -** Microsoft released Windows 98 (codenamed Memphis) on June 25, 1998. It included new hardware drivers and the FAT32 file system that supports disk partitions greater than 2 GB (initially included in Windows 95 OSR2)

**Microsoft Windows 2000 -** Microsoft released Windows 2000 in February 2000. It has the Windows NT version number 5.0. Windows 2000 has had four official service packs.

**Windows ME -** Microsoft launched a sequel to Windows 98 called Windows ME in September 2000, short for "Millennium Version" It was Microsoft's last operating system based on a DOS.

**Windows XP -** Microsoft unveiled Windows XP (codenamed "Whistler") on October 25, 2001. Finally, the merging of the Windows NT/2000 and Windows 95/98 / Me lines was achieved with Windows XP. Windows XP uses the Windows NT 5.1 kernel to replace the aging 16/32-bit branch, marking the entry of the Windows NT core to the consumer market.

**Windows Server 2003 -** Microsoft released Windows Server 2003 on April 25, 2003, a notable upgrade to Windows 2000 Server which includes several new security features, a new "Manage Your Server" wizard which simplifies the configuration of a system for different roles and enhances performance.

**Windows Server 2003 R2 -** On December 6, 2005, Windows Server 2003 R2, an upgrade to Windows Server 2003, was released to development. This is sold on two CDs, with one CD being the 2003 Windows Server SP1 DVD. The other DVD contains other potentially installable Windows Server 2003 features.

**Windows Vista -** Disseminated to industry consumers on 8 November 2006-Product models followed on 30 January 2007. Windows Vista intended to improve security by introducing a new restricted user mode named User Account Control, replacing Windows XP's "administrator by default" philosophy.

**Windows Server 2008 -** Windows Server 2008 was initially known as Windows Server Codename "Longhorn," launched on February 27, 2008. Windows Server 2008 builds on the advances in technology and security first introduced with Windows Vista, and is considerably more modular than its predecessor, Windows Server 2003.

**Windows 7 -** codenamed Vienna, originally Blackcomb is a personal computer operating system produced by Microsoft. It is part of the Windows NT operating system series. Windows 7 was released for manufacturing on 22 July 2009 and was commercially available on 22 October 2009, less than three years after its predecessor, Windows Vista, was released.

**Windows Server 2008 R2 -** Windows Server 2008 R2 is Microsoft-manufactured server operating system. It was released in development (RTM) on 22 July 2009 and launched on 22 October 2009.

**Windows 8 and Windows Server 2012 -** Microsoft released Windows 8 to the public on 26th October 2012. One edition, Windows RT, runs with mobile, 32-bit ARM (ARMv7) processors on some system-on-a-chip devices. Windows 8 features a redesigned user interface, designed to make using Windows easier for users of the touchscreen. Both Windows 8 Release Preview and Windows Server 2012 Release Candidate came out on May 31, 2012. Product development on Windows 8 was

completed on August 1, 2012, and released on the same day for manufacture. On 4 September 2012, Windows Server 2012 went on sale to the public. Windows 8 went on sale on 26th October 2012.

**Windows 8.1 and Windows Server 2012 R2 -** On 17 October 2013 Windows 8.1 and Windows Server 2012 R2 were released.

**Windows 10 -** Windows 10 is the Microsoft Windows operating system currently being released. It was unveiled on 30th August 2014 and released on 29th July 2015. It was distributed free of charge to owners of Windows 7 and 8.1 for a year after publication.

**Windows Server 2016 -** Windows Server 2016 is a cloud operating system developed by Microsoft as part of the operating system family Windows NT, developed at the same time as Windows 10. Along with the first technical demo of System Core, the first early release edition (Software Demo) became available on 1 October 2014. Windows Server 2016 was released at Microsoft's Ignite conference on September 6, 2016, and generally became available on October 12, 2016.

**Windows Server 2019 -** Windows Server 2019 was announced on March 20, 2018 and the first Preview version of Windows Insider was released on the same day. It was released on 2 October 2018, for general availability.

- ## Apple Mac OS X

***Image 1.2*** Apple MAC OS X View

**Mac OS** is a line of Apple-made operating systems. This comes preloaded on all new Macintosh, or Macs computers. Both the new releases are known as OS X (pronounced O-S Ten), and the special models include El Capitan (released in 2015), Yosemite (2014), Mavericks (2013), Mountain Lion (2012), Lion and (2011).

## History version of Mac OS

**Kodiak -** Apple launched a $29.95 "preview" version of Mac OS X (internally codenamed Kodiak) on September 13, 2000 to get consumers feedback.

**Cheetah -** Apple released Mac OS X 10.0 on March 24, 2001 (internally codenamed Cheetah).

**Puma -** Mac OS X 10.1 (internally codenamed Puma) was released later that year on September 25, 2001.

**Jaguar -** Apple followed up with Mac OS X on 23 August 2002 10.2 Jaguar, the first version to use the code word in the branding.

**Panther -** Panther Mac OS X 10.3 was released on 24 October 2003.

**Tiger -** On 29 April 2005 Mac OS X 10.4 Tiger was released. Apple said Tiger featured over 200 new apps.

**Leopard -** On 26 October 2007 Mac OS X 10.5 Leopard was released. It was named "Mac OS X's biggest redesign" by Apple.

**Snow Leopard -** Mac OS X 10.6 Snow Leopard was released August 28, 2009, the latest version to be released on disk.

**Lion -** On 20 July 2011 Mac OS X 10.7 Lion was released. It introduced Apple's iOS innovations, such as an conveniently navigable view of available applications (Launchpad) and (more use of) multi-touch gestures, to the Mac.

**Mountain Lion -** OS X 10.8 Released on July 25, 2012, Mountain Lion. This integrates some functionality found in iOS 5, including Game Center, iMessage support in the latest Messages Messaging system and Reminders as a dedicated to-do list function from iCal (which is called Calendar, as the iOS app).

**Mavericks** - OS X 10.9 Mavericks was released on October 22, 2013, as a free update through the Mac App Store worldwide.

**Yosemite -** On 16 October 2014, OS X Yosemite was released to the general public as a free update via the worldwide Mac App Store. This featured a massive user interface redesign, replacing skeuomorphism with flat graphic design, and distorted translucency effects, following the iOS 7 implemented aesthetic.

**El Capitan -** OS X El Capitan was announced during the keynote speech of the WWDC on 8 June 2015. It was released as a closed beta in July, and made publicly accessible on September 30th, 2015. Apple described this update as not new features but "Refinements to the Mac Experience" and "Improvements to Machine Performance."

**Sierra -** On 13 June 2016, macOS Sierra was unveiled at the keynote address of the WWDC. This was officially released on 20 September 2016.

**High Sierra -** On 25 September 2017 it was released to the public. Unlike OS X El Capitan and OS X Mountain Lion, High Sierra is a streamlined upgrade with a few new additions that are apparent to a user, including Safari, Photos and Mail upgrades, among other improvements.

**Mojave** - This was released on 24 September 2018. The upgrade added a system-wide dark mode and some additional features, such as the Apple News, were removed from iOS.

**Catalina** - It is the seventeenth major update of macOS, Apple's Macintosh application desktop operating system. It is the heir to macOS Mojave which was announced on 3 June 2019 at WWDC. Catalina is the first macOS update to support exclusively 64-bit applications.

**Big Sur** - In the WWDC keynote address, revealed on 22 June 2020. The main version number is updated, making it macOS 11.0 for the first time in a macOS update. It introduces into the system ARM support, brand new icons, and GUI.

## ➢ Linux

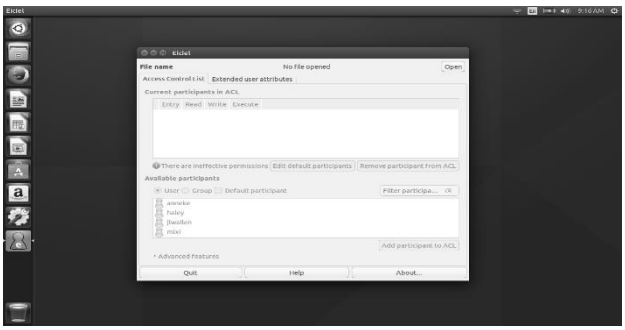


***Image 1.3*** Linux GUI View

**Linux** (pronounce LINN-ux) is a family of open source operating systems, meaning that anyone in the world can change and spread them. That is somewhat different from proprietary applications such as Windows, which only the corporation that owns it (Microsoft) can change. The benefits of Linux are that it's free, so you can pick from several different distributions (or versions.

**Linux** is an operating system much like Windows, iOS and Mac OS. In reality, the Linux operating system is operated by one of the most popular systems on the world, Android.

**Linux** takes its name from Linus Torvalds, who developed the 1991 Linux kernel. The kernel is the programming of a machine and is the core component of an operating system.

In fact, Ubuntu is licensed under an open source license. Those primary tenants follow open source:

- The right to administer the programme, for whatever reason.
- The ability to learn and change how the system works, to make it do what you want.
- The right to redistribute copies for the neighbor to support.
- The right to make copies of the updated versions accessible to anyone.

Linux has different variants to fit each form of customer. Including novice users to hard-core users, you'll find a Linux "flavor" that suits your needs. Both variants are called distributions (or "distros" in the short form). Nearly any Linux distribution

can be downloaded free of charge, burned to disk (or USB thumb drive) and installed (on as many computers as you wish).

## Linux distributions

**LINUX MINT** - Ubuntu-based distribution which aims to provide a classic desktop environment with various handy, personalized tools and optional multimedia out-of-the-box support.

**MANJARO** - Manjaro Linux is an operating system based on Arch Linux, which is fast, user-friendly and desktop oriented. Main features include easy installation process, automated hardware discovery, reliable roll-release style, multiple kernel installation support, special Bash scripts for graphics driver management and robust screen configurability.

**DEBIAN** - Debian comes with over 50,000 packages (precompiled applications bundled in a pleasant format to enable quick deployment on your machine)-all free. It's like a hill, a little. The kernel is at the heart

**UBUNTU** - Ubuntu is a full Linux desktop operating system, available free of charge with community and technical support. "Ubuntu "is an ancient African term, meaning" humanity to all. The Ubuntu distribution is taking Ubuntu's spirit into the realm of tech.

**ANTERGOS** - Is a new, streamlined and powerful Arch Linux-based operating system. It started life under the name Cinnarch, merging the Cinnamon desktop with the Arch Linux distribution, but the project moved away from its original aims and now provides a variety of multiple desktops, including GNOME 3 (default), Cinnamon, Razor-qt and Xfce.

**SOLUS** - Is a scratch optimized Linux distribution. It uses a forked version of the PiSi package manager, maintained within Solus as "eopkg," and an in-house built custom workspace environment, called "Budgie."

**FEDORA** - The Linux kernel (formerly Fedora Core) is developed by the community-supported Fedora Project and operated by Red Hat. Fedora includes software that is licensed under a free, open-source license and strives to be at the forefront of these technology. Fedora has a reputation for focusing on creativity, early adoption of emerging technology and close cooperation with upstream Linux communities.

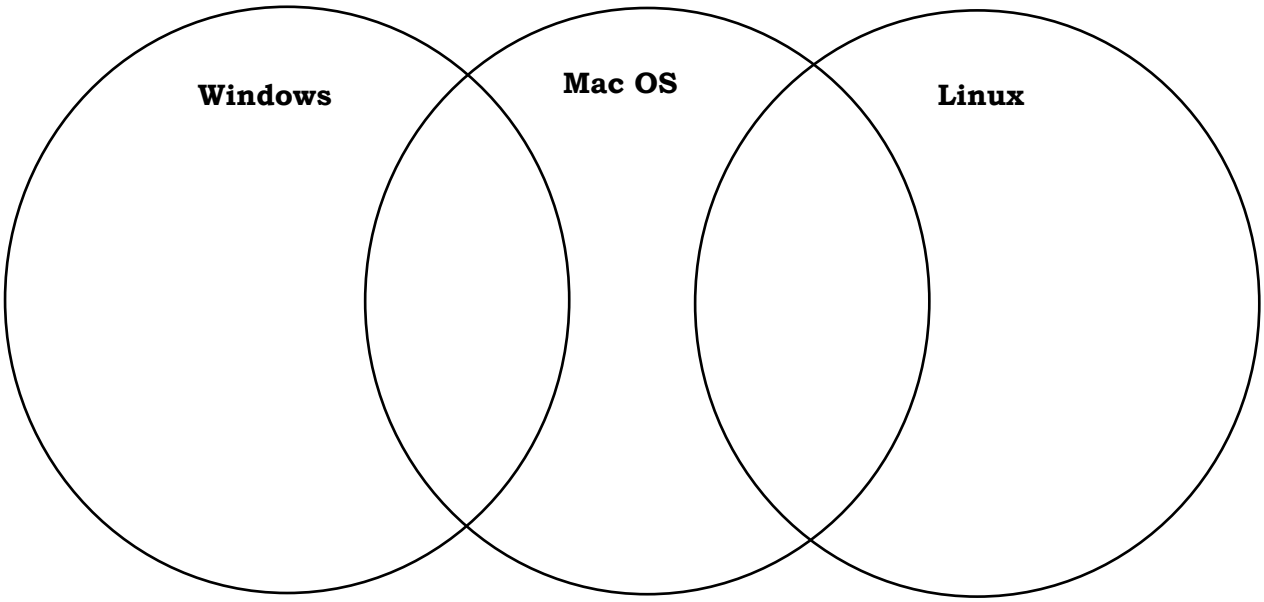
**ELEMENTARY OS** - A Linux Server based on Ubuntu. Many of its most notable features include Pantheon 's unique desktop environment and other custom applications including images, songs, videos, calendar, browser, data, and more.

**OPENSUSE** - The openSUSE initiative is a community-sponsored platform for SUSE Linux and other organizations. The software encourages the use of Ubuntu everywhere and gives secure, fast access to openSUSE, a complete Linux

distribution.

# ACTIVITIES

**A. Direction:** Write down the differences of Computer operating system in the Venn Diagram.




**B. Directions:** Create an infographic showing the version history of different Operating System.

# WRAP-UP

This module talked about Operating Systems. Specifically, it discussed the different versions of Windows OS, MAC OS and Linux OS.

In addition, Operating system (OS) of your computer manages all of the computer's software and hardware. Most of the time, there are several different computer programs running simultaneously and all of them need to access the central processing unit (CPU), memory, and storage of your computer. All of this is managed by the operating system to ensure each application gets what it wants.

Now, as learners, aside from the several points mentioned above. You can use the blank spaces provided in the next page to answer and react to the following cases or scenarios:

- What are the challenges you encountered in learning this module?

  _____

  _____

  _____

# VALUING

**DIRECTION:** Read and answer the following questions carefully in two to three sentences each number.

1. What do you think are the importance of learning the history of Computer Operating System?

   _____

   _____

   _____

2. How will you use the knowledge you acquired about Computer Operating System?

   _____

   _____

   _____

3. Cite a situation in which you can apply the knowledge of understanding Computer Operating System.

   _____

   _____

   _____

# POSTTEST

**I.** **Direction:** Write *T* on the line if the statement is correct and *F* if the statement is wrong.

___1. Computer's operating system (OS) manages all of the software and hardware on the computer.
___2. Microsoft created the Windows operating system in the mid-1970s.
___3. Windows NT is a family of operating systems produced by Microsoft, the first version of which was released in July 1993.
___4. Windows 95 (codenamed Chicago) is a consumer- oriented operating system developed by Microsoft.
___5. Windows 7 (codenamed Vienna, formerly Blackcomb is a personal computer operating system developed by Microsoft.
___6. On September 25, 2002, Mac OS X 10.1 (internally codenamed Puma) was released.
___7. Mac OS X 10.7 Snow Leopard was released on August 28, 2009, the last version to be available on disc.
___8. OS X 10.9 Mavericks was released on October 22, 2013, as a free update through the Mac App Store worldwide.
___9. Linux takes its name from Linus Torvalds, who developed the 1991 Linux kernel.
___10. Manjaro Linux is an operating system based on Arch Linux, which is fast, user-friendly and desktop oriented.

**II.** **Direction:** Choose the letter of the correct answer.

1. It manages the computer's memory, processes, and all of its software and hardware.
   a. System Software b. Application Software c. Operating System
2. the computer code that is the central part of an operating system.
   a. Kernel b. CPU c. Processor
3. the largest update of Mac OS X".
   a. Leopard b. Lion c. Puma
4. Codenamed Vienna, formerly Blackcomb is a personal computer operating system developed by Microsoft.
   a. Windows 8 b. Windows 7 c. Windows 10
5. The first independent version of Microsoft Windows.
   a. Windows 1.0 b. Windows 2.0 c. Windows 3.0

# KEY TO CORRECTION

**Postest Key to Correction**

| | |
|---|---|
| I. | II. |
| 1. T | 1. C |
| 2. F | 2. A |
| 3. T | 3. A |
| 4. T | 4. B |
| 5. T | 5. A |
| 6. F | |
| 7. F | |
| 8. T | |
| 9. T | |
| 10. T | |

**Activities Key to Correction**

1. Linux Mint, Manjaro, Debian, Ubuntu, Antegros,Solus,Fedora
2. Panther, Jaguar, Tiger, Leopard, Lion, Puma, Big Sur
3. Microsoft Windows, Mac OS X, Linux
4. Win 7, Win 8, Win 8.1, Win 10, Win ME, Win Vista

**Pretest Key to Correction**

1. Operating System
2. Microsoft
3. Windows 1.0
4. Windows ME
5. Mac OS X
6. Leopard
7. Linux
8. Manjaru
9. SOLUS
10. UBUNTU

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